

## 1 Announcements

Anonymous survey: <https://forms.gle/RrTDwdwF5S35f8cT8>

## 2 Discussion Outline

- (1) Exponential and logarithmic functions
- (2) Worksheet 2

## 3 Notes

### 3.1 Review from Math 31A:

**Theorem 3.1** (Chain Rule).

$$(f \circ g)'(x) = f'(g(x))g'(x)$$

**Theorem 3.2** (Substitution Theorem).

$$\int f(u(x))u'(x) = \int f(u)du$$

**Lemma 3.1** (Interchanging limits and continuous functions). If  $g$  is continuous at  $c$  and  $f$  is continuous at  $g(c)$ , then  $f \circ g$  is continuous at  $c$ , which is to say

$$\lim_{x \rightarrow c} f(g(x)) = f(g(c)) = f(\lim_{x \rightarrow c} g(x))$$

### 3.2 Exponential function

**Definition 3.1** (Exponential functions). An **exponential function** is of the form  $f(x) = b^x$  for some  $b > 0$ . By convention  $b \neq 1$ . We call  $b$  the **base**.

**Proposition 3.1** (Derivative of exponential function). If  $f(x) = b^x$ , for some  $b > 0$ , then

$$\frac{d}{dx} f(x) = \lim_{h \rightarrow 0} \left( \frac{b^h - 1}{h} \right) b^x$$

where  $\lim_{h \rightarrow 0} \left( \frac{b^h - 1}{h} \right)$  exists.

**Proposition 3.2 (Differentiability implies continuity).** If  $f$  is differentiable at  $x = c$ , then it is continuous at  $x = c$ .

*Proof.* We have

$$\lim_{x \rightarrow c} (f(x) - f(c)) = \lim_{x \rightarrow c} \left( \frac{f(x) - f(c)}{x - c} \cdot (x - c) \right) = f'(c) \cdot 0 = 0$$

□

**Proposition 3.3 (Properties of exponential functions).** Let  $f(x) = b^x$  be an exponential function. Then  $f$  satisfies:

- (i)  $f$  is continuous everywhere.
- (ii)  $f(x) = b^x > 0$
- (iii)  $\text{range}(f) = (0, +\infty)$
- (iv) If  $b > 1$ , then  $f$  is increasing. If  $0 < b < 1$ , then  $f$  is decreasing.

**Definition 3.2 (Euler's number).** We define  $e$  to be the unique real positive number satisfying

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

**Remark 3.1.** We have

$$\frac{d}{dx} e^x = e^x$$

**Remark 3.2.** There are other equivalent definitions of  $e$ . For example,  $e$  can be written as a series of numbers, which we will see later in the course.

### 3.3 Logarithmic function

**Definition 3.3 (Inverse function).** Let  $f$  be some function. An **inverse** of  $f$  is a function  $g$  such that

$$f(g(x)) = x \quad \text{and} \quad g(f(x)) = x$$

If it exists, it is unique and we write  $f^{-1}$ .

**Definition 3.4 (Logarithmic functions).** A **logarithmic function** with base  $b > 0$  is the inverse function of  $b^x$ . We write it as  $\log_b(x)$ .

**Theorem 3.3 (Derivative of inverse).** Suppose  $f$  is differentiable and one-to-one with inverse  $g(x)$ . If  $b$  is in the domain of  $g$  with  $f'(g(b)) \neq 0$ , then  $g'(b)$  exists and

$$g'(b) = \frac{1}{f'(g(b))}$$

**Theorem 3.4 (Derivative of logarithmic functions).**

$$\frac{d}{dx} \log_b(x) = \frac{1}{x \ln(b)}$$

*Proof.* Change of base formula gives

$$\log_b(x) = \frac{\ln(x)}{\ln(b)}.$$

Then we have

$$\frac{d}{dx} \log_b(x) = \frac{1}{\ln(b)} \frac{d}{dx} \ln(x) = \frac{1}{\ln(b)} \frac{1}{x}$$

□

**Proposition 3.4 (Properties of logarithmic functions).** Let  $f(x) = \log_b(x)$  be a logarithmic function. Then  $f$  satisfies:

- (i)  $f$  is continuous (on its domain)
- (ii)  $f(1) = 0$
- (iii)  $\text{range}(f) = (-\infty, +\infty)$
- (iv) If  $b > 1$ , then  $f$  is increasing. If  $0 < b < 1$ , then  $f$  is decreasing.

**Remark 3.3.** Combining the derivative of  $\log_b(x)$  with the derivative of the inverse gives

$$\frac{d}{dx} b^x = \ln(b) b^x \quad \text{so} \quad \lim_{h \rightarrow 0} \frac{b^h - 1}{h} = \ln(b)$$

Remembering this derivative is easy if we note that

$$b^x = e^{x \ln(b)}$$

and use chain rule.

**Corollary 3.1 (Antiderivative of  $x^{-1}$ ).** An antiderivative of  $x^{-1}$  over its domain is  $\ln|x|$ , which is to say:

$$\int \frac{1}{x} dx = \ln|x| + C$$